

Testing Guide

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This is the **authoritative test-type catalogue** for qBittorrent-Fixed. Every testable module in the repo must have coverage in every applicable row below. The catalogue is derived from Part C of the completion-initiative plan (docs/superpowers/plans/2026-04-19-completion-initiative.md).

Test types

#	Type	Framework	Directory	How to run	Coverage / report lands in
1	Unit (Python)	pytest + pytest-asyncio	tests/unit/	pytest tests/unit/ -v --import-mode=importlib	htmlcov/, coverage.xml
2	Unit (Frontend)	Vitest + Angular Testing Library	frontend/src/**/*.spec.ts	cd frontend && ng test	frontend/coverage/
3	Integration (API)	pytest + httpx.AsyncClient	tests/integration/	pytest tests/integration/ -v --import-mode=importlib	htmlcov/, coverage.xml
4	Integration (Plugin â†”i,Ž nova3)	pytest + verpy (record/replay)	tests/integration/plugins/ (pending Phase 4)	pytest tests/integration/plugins/ -v	htmlcov/
5	End-to-end (browser)	Playwright (Python)	tests/e2e/	pytest tests/e2e/ -v	artifacts/e2e/
6	End-to-end (API flow)	pytest + live compose	tests/e2e/ (fixture-gated)	pytest tests/e2e/ -v --import-mode=importlib	htmlcov/
7	Contract (OpenAPI)	schemathesis	tests/contract/	schemathesis run http://localhost:7187/openapi.json	artifacts/contract/
8	Property-based hypothesis		tests/property/ (pending)	pytest tests/property/ -v	htmlcov/
9	Fuzz	hypothesis + atheris (optional)	tests/fuzz/ (pending)	pytest tests/fuzz/ -v	artifacts/fuzz/
10	Mutation	mutmut	runs against download-proxy/src/ (pending)	mutmut run	html/ mutmut.html
11	Security	pytest + bandit + semgrep rules	tests/security/	pytest tests/security/ -v	

#	Type	Framework	Directory	How to run	Coverage / report lands in htmlcov/ + artifacts/ scans/
12	Performance (latency)	pytest-benchmark	tests/ performance/ tests/ benchmark/	pytest tests/ performance/ tests/benchmark/ -v --benchmark- only	.benchmarks/
13	Load (throughput)	Locust	tests/load/ (pending Phase 6)	locust -f tests/ load/ locustfile.py	artifacts/ load/
14	Stress (overload)	Locust + chaos probes	tests/stress/	pytest tests/ stress/ -v	artifacts/ stress/
15	Chaos (fault injection)	toxiproxy + pytest	tests/chaos/ (pending)	pytest tests/ chaos/ -v	artifacts/ chaos/
16	Concurrency (race)	pytest-randomly + pytest-repeat + asyncio ops	tests/ concurrency/ (pending)	pytest tests/ concurrency/ -v --count=10	htmlcov/
17	Memory leak	tracemalloc + pytest + objgraph	tests/memory/ (pending)	pytest tests/ memory/ -v	artifacts/ memory/
18	Deadlock / timeout	pytest-timeout (hard cap all tests)	global	pytest -v (30 s timeout is the default in pyproject.toml)	test log
19	Smoke	pytest	tests/smoke/ (pending)	pytest tests/ smoke/ -v	htmlcov/
20	Monitoring / metrics	pytest + Prometheus scrape asserts	tests/ observability/ (pending)	pytest tests/ observability/ -v	htmlcov/
21	Infra / compose	pytest-docker + compose-config validator	tests/infra/ (pending)	pytest tests/ infra/ -v	htmlcov/
22	Accessibility	Playwright + axe- core	tests/a11y/ (pending)	pytest tests/ a11y/ -v	artifacts/ a11y/
23	Visual regression	Playwright snapshots	tests/visual/ (pending)	pytest tests/ visual/ -- update-snapshots	tests/visual/ __snapshots__
24	Documentation link check	pytest + linkchecker	tests/docs/ (pending)	pytest tests/ docs/ -v	artifacts/ docs/
25	Type-check (static)	mypy --strict + tsc --noEmit	CI job	mypy download- proxy/src plugins/cd frontend && tsc --noEmit	CI log
26	Lint (static)	ruff + eslint + shellcheck + hadolint + yamllint	CI job	ruff check ./cd frontend && ng lint/shellcheck *.sh	CI log
27			CI job		

#	Type	Framework	Directory	How to run	Coverage / report lands in
	Dependency audit	pip-audit + npm audit + Snyk + Trivy		pip-audit -r download-proxy/requirements.txt / snyk test	artifacts/scans/
28	SAST	Semgrep + Bandit + SonarQube	CI job	semgrep scan -- config=auto / bandit -r download-proxy/src	artifacts/scans/
29	Secret scan	Gitleaks + TruffleHog	CI job	gitleaks detect	artifacts/scans/
30	License compliance	pip-licenses + license-checker	CI job	pip-licenses / cd frontend && license-checker	artifacts/licenses/

Rows marked *pending* are scaffolded by the completion-initiative plan phases (see the plan document for the phase that ships each one).

Where tests live

- **Top-level tests** / “all merge-service tests, *not* download-proxy/tests/ (which does not exist).
- `frontend/src/**/*.spec.ts` co-located Angular unit tests.
- **CI jobs** “`.github/workflows/{syntax,unit,integration,nightly,security}.yaml`.

Service-availability fixtures

Since Phase 0 the following fixtures **start the compose stack** rather than skip when services are down (converted 71 runtime skips to fixture-gated executions, per commit 55d29ce):

Fixture	Starts	Asserts
<code>merge_service_live</code>	<code>qbittorrent-proxy</code>	GET http://localhost:7187/health â†’ 200
<code>qbittorrent_live</code>	<code>qbittorrent</code>	GET http://localhost:7186/ â†’ 200
<code>webui_bridge_live</code>	<code>webui-bridge.py</code>	GET http://localhost:7188/health â†’ 200
<code>all_services_live</code>	all three	all of the above

Defined in `tests/fixtures/services.py` and wired into `tests/conftest.py`.

How to add a new test (TDD cadence)

The CLAUDE.md critical constraint is non-negotiable:

1. **RED** “write the failing test first. Name it after the behaviour, not the implementation. Put it in the directory that matches the test type from the catalogue above.
2. **Watch it fail** “run the test in isolation (`pytest tests/unit/test_new_thing.py::test_specific_case -v`) and confirm the failure reason is the one you want (NOT an import error, NOT a fixture error).
3. **GREEN** “write the minimum production code to make the test pass. No refactors yet.
4. **Verify** “run the whole affected test directory to prove nothing else regressed.
5. **Rebuild-reboot** “if the change touched container code, follow the CLAUDE.md REBUILD AND REBOOT constraint:

```
./stop.sh
podman exec qbittorrent-proxy find / -name __pycache__ -type d -exec rm -rf {} \;
./start.sh -p
```

Curl-check the served content matches the committed code.

6. **Commit** “one behaviour per commit. Commit message follows `<type> (phase-N) : <subject>` when part of a phased plan.

Coverage gate

- `pyproject.toml` contains `[tool.coverage.report]` with `fail_under` set to the current gate (Phase 0 starts low; Phase 5 raises it module-by-module to 100 %).
- The baseline zero point is captured in [COVERAGE_BASELINE.md](#).
- HTML reports land in `htmlcov/`; XML (for SonarQube) in `coverage.xml`.

CI entry points

- **Local full pipeline** “`./ci.sh`
- **Quick loop** “`./ci.sh --quick`
- **Hardcoded-podman runner** “`./run-all-tests.sh` (note: hardcodes podman, will fail on docker-only systems “see CLAUDE.md gotcha)
- **Single-file** “`./test.sh --plugin <name> or ./test.sh --full`

Per-tracker search stats

Tests that cover `TrackerSearchStat` and the `tracker_started / tracker_completed` SSE events live across four suites:

Test file	Type	What it pins
<code>tests/unit/merge_service/test_tracker_stats.py</code>	Unit	<code>TrackerSearchStat</code> defaults, status transitions (pending “running “success / empty / error / timeout),

Test file	Type	What it pins
frontend/src/app/services/theme.service.spec.ts	Unit (Vitest)	Sass variables; dashboard template includes <app-theme-picker>; TOKEN_CSS_VAR map covers every var. System-dark detection, stored-state hydration, unknown-id fallback, setPalette writes all 15 tokens with exact values to documentElement.style, toggleMode flips signal + data-mode + color-scheme, prefers-color-scheme change listener respects modeIsUserChosen flag. One menu item per palette, swatches render with accent/contrast/bg/text colours, clicking an item calls setPalette, clicking outside closes the menu, active palette has .active + aria-checked="true".
frontend/src/app/components/theme-picker/theme-picker.component.spec.ts	Unit (Vitest)	Live dashboard at :7187: picking a new palette changes --color-accent (+3 others) on <html> to the catalogue value AND writes qbitt.theme into localStorage. Skips with a loud message when the served bundle predates the theme-picker.
tests/e2e/test_theme_runtime.py	End-to-end (Playwright)	

Whenever the TrackerSearchStat shape changes, run ./scripts/freeze-openapi.sh to refresh docs/api/openapi.json so the contract test test_frozen_and_live_have_same_schemas stays green.

Gotchas

- pytest needs --import-mode=importlib for the merge service “the project is not a Python package (no __init__.py at the root).
- Integration tests with requests.get("http://localhost:7187/").ok guards are **obsolete** since Phase 0 “use the live fixtures instead. tests/unit/test_no_runtime_service_skips.py enforces this.
- pytest-timeout hard-caps every test at 30 seconds. Stress and load tests that legitimately run longer must override with @pytest.mark.timeout(...).

- The CI workflow is **no longer manual-only** since Phase 0.4 “push and PR triggers the syntax / unit / integration workflows. Only the `nightly.yml` and `security.yml` are scheduled.